

WEEK-9

Task-1: To Implement how to List Active Jobs, Display Status Information, Retrieve Job [Ds, Format Output, Update State Information, Test Job Listings

```
pavani@DESKTOP-380LEIN:~/skill$ ./myshell
-bash: ./myshell: No such file or directory
pavani@DESKTOP-380LEIN:~/skill$ sleep 30 &
p 40 &
sleep 50 &[1] 2851
pavani@DESKTOP-380LEIN:~/skill$ sleep 40 &
[2] 2861
pavani@DESKTOP-380LEIN:~/skill$ sleep 50 &
[3] 2871
pavani@DESKTOP-380LEIN:~/skill$ jobs
[1]    Running                  sleep 30 &
[2]-  Running                  sleep 40 &
[3]+  Running                  sleep 50 &
pavani@DESKTOP-380LEIN:~/skill$ sleep 5 &
[4] 2890
pavani@DESKTOP-380LEIN:~/skill$ jobs
[1]    Done                     sleep 30
[2]    Running                  sleep 40 &
[3]-  Running                  sleep 50 &
[4]+  Done                     sleep 5
pavani@DESKTOP-380LEIN:~/skill$
```

Report : The shell was extended to maintain and display active background jobs. Each job stores information such as job ID, process ID, command, and current state. The jobs command was tested with multiple background processes. Job states were updated when processes completed or changed state.

Task-2: To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

```

pavani@DESKTOP-380LEIN: ~/skill
pavani@DESKTOP-380LEIN:~$ cd skill
pavani@DESKTOP-380LEIN:~/skill$ ./myshell
-bash: ./myshell: No such file or directory
pavani@DESKTOP-380LEIN:~/skill$ ls
append      background  combined    combined.txt  complex.c    error.txt    jobtable.c    stderr
append.c    background.c  combined.c  complex      data.txt    jobtable     result.txt    stderr.c
pavani@DESKTOP-380LEIN:~/skill$ grep -l "int main" *.c
append.c
background.c
combined.c
complex.c
jobtable.c
stderr.c
pavani@DESKTOP-380LEIN:~/skill$ sleep 30 &
[1] 2667
pavani@DESKTOP-380LEIN:~/skill$ jobs
[1]+  Running                  sleep 30 &
pavani@DESKTOP-380LEIN:~/skill$ fg 1
-bash: fg: warning: 1: job specification requires leading `%`
-bash: fg: job has terminated
[1]+  Done                      sleep 30
pavani@DESKTOP-380LEIN:~/skill$ tcsetpgrp()
> waitpid()
-bash: syntax error near unexpected token `waitpid'
pavani@DESKTOP-380LEIN:~/skill$ tcsetpgrp()
> waitpid()
-bash: syntax error near unexpected token `waitpid'
)pavani@DESKTOP-380LEIN:~/skill$ kill()
>

```

Report : Foreground job switching was implemented by identifying a job from the job table. Terminal control was transferred to the selected process group using terminal process-group management.

The shell waited for the foreground process and regained terminal control afterward.

Job status was updated based on whether the process completed or was stopped.

Task-3: To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior

```

pavani@DESKTOP-380LEIN:~/skill$ ./myshell
-bash: ./myshell: No such file or directory
pavani@DESKTOP-380LEIN:~/skill$ sleep 30
^C
pavani@DESKTOP-380LEIN:~/skill$ sleep 30
^Z
[1]+  Stopped                  sleep 30
pavani@DESKTOP-380LEIN:~/skill$ jobs
[1]+  Stopped                  sleep 30
pavani@DESKTOP-380LEIN:~/skill$ ps -o pid,ppid,pgid,sid,stat,cmd
  PID   PPID    PGID     SID STAT  CMD
  2574   2570    2574     2574 Ss    -bash
  2940   2574    2940     2574 T     sleep 30
  2960   2574    2960     2574 R+    ps -o pid,ppid,pgid,sid,stat,cmd
pavani@DESKTOP-380LEIN:~/skill$

```

Report: Signal delivery and terminal-generated signals were studied using foreground processes and process groups.

Ctrl+C was used to generate SIGINT, while Ctrl+Z generated SIGTSTP.

Process IDs, process groups, and session information were examined using ps.

Signal masks and signal handling behavior were also investigated.

Task-4: To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verifr Execution, Test Background Recovery

```

[1]+  Stopped                  sleep 30
pavani@DESKTOP-380LEIN:~/skill$ ps -o pid,ppid,pgid,sid,stat,cmd
  PID   PPID    PGID     SID STAT  CMD
  2574   2570    2574     2574 Ss    -bash
  2940   2574    2940     2574 T     sleep 30
  2960   2574    2960     2574 R+    ps -o pid,ppid,pgid,sid,stat,cmd
pavani@DESKTOP-380LEIN:~/skill$ bg 1
-bash: bg: warning: 1: job specification requires leading `%'
[1]+ sleep 30 &
[1]+  Done                    sleep 30
pavani@DESKTOP-380LEIN:~/skill$ jobs
pavani@DESKTOP-380LEIN:~/skill$ fg 1
-bash: fg: warning: 1: job specification requires leading `%'
-bash: fg: 1: no such job
pavani@DESKTOP-380LEIN:~/skill$

```

Report: Stopped jobs were resumed using the SIGCONT signal.

The shell updated the job table from Stopped to Running after recovery.

Background recovery was tested using the bg command, while fg was used to return jobs to the foreground.

The experiment demonstrated how signals and process groups are used to implement shell job control.

